Assignment 8: Task List, Milestones, Timeline, and Effort Matrix

Verified Student Sublease Marketplace | CS5001 Senior Design I | Covers official Assignment #5 (Task Lists) and #6 (Milestones, Timeline, and Effort Matrix)

Item	Detail
Team members	Danvanth Vanganuri, Jovan Jose Asker Fredy, Mohamed Aidja
Goal	Let UC students fill or find term-length subleases safely. Only verified @mail.uc.edu students can use it. Listings match exact co-op dates. Discriminatory listings are held for review. Property managers approve or reject before move-in.
Built from	Project-Description.md, User_Stories, D0 (C1 to C8, I1 to I15), D1 (C3, C5), D2 (C1 to C8, G1 to G3, open questions GQ-01 to GQ-08).
Repository	Tasklist.md in the repository root. Plan sources in Planning/ (plan.py, build.py, CSV tables, Gantt.png).
Contents	1. Task List 2. Milestones List 3. Table 1: Timeline (+ Gantt) 4. Table 2: Effort Matrix 5. Work Breakdown Structure 6. Risk Register 7. Consistency check

How we made this plan

- **Tasks** come from the modules in D0 to D2. Each task is one module (or one part of a module) and one owner.
- **Owners** follow the D0/D2 component owners. Jovan: C3, C4, C5, G2. Danvanth: C2, C7, C8, G1, E1. Mohamed: C1, C6, G3.
- **Estimates** use three numbers per task: optimistic (O), most likely (M), and pessimistic (P). Expected hours = (O + 4M + P) / 6.
- **Time per person**: about 10 hours a week in Fall 2026 and 14 hours a week in Spring 2027 (CS 5002).
- **Buffer**: every task gets 20% more time. Each major milestone also has a protected buffer block. No new work starts in a block. It is only for fixing and joining parts.
- **Calendar**: no work on Thanksgiving, in the last fall week, in fall exam week, in winter break, or in spring break. Spring dates assume classes start Jan 11, 2027 and spring break is Mar 15 to 19. We re-check them at M5.

1. Task List (Assignment #5)

18 tasks. Six per member. Each task starts with a verb, names its Design Diagram module, and has one owner. Each fits in one or two weeks. The same list is in Tasklist.md.

ID	Task statement	Design Diagram module	Owner (primary)	Weeks
T1	Develop the project skeleton, the CI pipeline, the G2 Configuration module, and the E3 database schema migrations.	G2 Configuration; E3 Managed SQL DB	Jovan Jose Asker Fredy	2
T2	Research and obtain the E1 Email Delivery API account, and confirm its free-tier limit, webhook signing, and idempotency keys.	E1 Email Delivery API (for C8)	Danvanth Vanganuri	1
T3	Design the C1 Web Client wireframes and the accessibility checklist for every user story screen.	C1 Web Client	Mohamed Aidja	2
T4	Specify the C4 Compliance Screener phrase list v1, including the Ohio R.C. 4112 protected classes.	C4 Compliance Screener	Jovan Jose Asker Fredy	1
T5	Develop the G3 Logging and audit module (log, audit record, and the /healthz check).	G3 Logging and audit	Mohamed Aidja	1
T6	Develop the G1 Authentication middleware and the signed session token.	G1 Authentication middleware	Danvanth Vanganuri	2
T7	Develop the C2 Identity Service for @mail.uc.edu login links, the 24-hour link expiry, and rate limits (I2, I10).	C2 Identity Service	Danvanth Vanganuri	2
T8	Develop the C4 Compliance Screener screen() function (I7) with unit tests for every phrase category.	C4 Compliance Screener	Jovan Jose Asker Fredy	1
T9	Develop the C3 Listing Service create, validate, hold, publish, and review endpoints (I3).	C3 Listing Service	Jovan Jose Asker Fredy	2
T10	Develop the C3 outbox relay and the C5 Search & Match index and date-coverage search (I8, I4).	C3 Listing Service; C5 Search & Match	Jovan Jose Asker Fredy	2
T11	Develop the C8 Notification Service queue, the E1 send with retries, the status webhook, and the suppression list (I10 to I13).	C8 Notification Service	Danvanth Vanganuri	2
T12	Develop the C1 Web Client screens for sign-in, create listing, and search (I1 to I4).	C1 Web Client	Mohamed Aidja	2
T13	Develop the C6 Messaging Service threads, messages, and alert limits (I5, I11).	C6 Messaging Service	Mohamed Aidja	2
T14	Develop the C7 Approval Service proposals, decisions, unit registry seed, and daily reminder job (I6, I9, I12).	C7 Approval Service	Danvanth Vanganuri	2
T15	Develop the C1 Web Client screens for messages, approval requests, and the review queue (I5, I6).	C1 Web Client	Mohamed Aidja	2
T16	Test the timing budgets and error paths for AC-01.1, AC-02.1, and AC-02.2 with automated end-to-end tests.	C3, C7, C8 (outbox and email path)	Danvanth Vanganuri	2
T17	Validate US-04 accessibility with axe-core in CI and a manual NVDA and VoiceOver pass.	C1 Web Client	Mohamed Aidja	2
T18	Validate the release candidate with a five-student user test, and document the results and the user guide.	Whole system (C1 to C8)	Jovan Jose Asker Fredy	2

Tasks per member: Jovan Jose Asker Fredy 6, Danvanth Vanganuri 6, Mohamed Aidja 6.

2. Milestones List (Assignment #6)

A milestone is a checked, dated result. Course deliverables are listed first. Completed ones are marked **Complete**. Internal milestones are every 1 to 3 weeks. The only longer gap is fall exams plus winter break, when no work is planned.

ID	Milestone	Type	What it shows	Date	How it is checked	Checked by	Status
M0a	Project description and Week 3 constraints	Course	The problem, the goal, the users, and the budget, hosting, privacy, and accessibility limits are written down.	Week 3 (Sep 2026)	Project-Description.md is in the repository and was graded.	Instructor	**Complete**
M0b	User Stories (US-01 to US-05, UC-01, UC-02, AC-01.1 to AC-02.2)	Course	Every feature has a user story, a use case, and acceptance criteria we can test.	Week 5 (Sep 2026)	User_Stories document is in the repository and was graded.	Instructor	**Complete**
M0c	Design D0: High-Level Design	Course	The system is split into C1 to C8 with 15 interfaces, two data flows, and a decision log.	Oct 2, 2026	D0 document and .drawio files are in Design_Diagrams.	Instructor	**Complete**
M0d	Design D1: Detailed Design of C3 and C5	Course	Data model, algorithms, and API for the Listing Service and Search & Match.	Week 7 (Oct 2026)	D1 document is in Design_Diagrams.	Instructor	**Complete**
M0e	Design D2 and peer review (consolidated Assignment #4 PDF)	Course	All components, glue modules G1 to G3, state machines, edge cases, and traceability are designed. Peer review found no issues.	Oct 14, 2026	Consolidated PDF submitted on Canvas by each member; D2 in Design_Diagrams.	Instructor and peer teams	**Complete**
M1	Assignment 8: Task list, milestones, timeline, effort matrix	Course	The design is turned into 18 owned tasks with dates, effort, and risks.	Oct 28, 2026	This PDF on Canvas; Tasklist.md in the repository root; sources in Planning/.	Instructor and advisor	**Planned**
M2	Development environment ready	Internal (major)	Code builds and deploys. A real email reaches a @mail.uc.edu inbox (not junk). The search date index works on our database.	Nov 13, 2026	Demo at the weekly meeting: CI is green on main, /healthz answers on Render, migrations run on Supabase, test email screenshot, btree_gist query runs.	Mohamed checks T1; Jovan checks T2 and T3	**Planned**
M3	Week 14 report (reproduces these tables)	Course	Progress against this plan is reported, with updated Table 1 and Table 2.	Nov 25, 2026	Report submitted on Canvas by each member.	Instructor and advisor	**Planned**
M4	Phase 1 done: sign-in base and screening rules	Internal (major)	G1 accepts a valid signed token and rejects a bad one. G3 writes log and audit rows. Phrase list v1 is approved.	Dec 4, 2026	G1 and G3 unit tests pass in CI; advisor signs off on the phrase list.	Advisor (phrase list); Jovan (tests)	**Planned**
M5	Spring re-kickoff and re-baseline	Internal	Every member's spring schedule and co-op status is confirmed. CS 5002 dates are added to the plan.	Jan 15, 2027	Updated Table 1 committed to Planning/; meeting notes list each member's status.	Whole team; advisor informed	**Planned**
M6	Walking skeleton	Internal (major)	A student signs in with an emailed link, creates a listing, and a listing with a flagged phrase is held for review.	Feb 5, 2027	Live demo on the deployed site, plus an automated end-to-end test in CI.	Advisor at the weekly check-in	**Planned**
M7	Core flows work end to end	Internal (major)	Search by co-op dates, messaging, and property-manager approval all work, with real emails.	Feb 26, 2027	Demo of Data Flow A and Data Flow B from D0 on the deployed site.	Advisor	**Planned**
M8	Feature complete	Internal (major)	All five user stories work and every acceptance criterion has a passing automated test.	Mar 12, 2027	CI report: all AC tests pass, zero axe-core violations, timing tests inside budget.	Advisor; Danvanth runs the report	**Planned**
M9	User test sessions done	Internal	Five UC students used the system and their problems are logged.	Mar 26, 2027	Signed-off test notes and an issue list in the repository.	Mohamed and Danvanth (session helpers)	**Planned**
M10	Release candidate v1.0	Internal (major)	User test issues are fixed, v1.0 is tagged and deployed, and the user guide is written.	Apr 9, 2027	Git tag v1.0; the deployed site passes the full test suite; user guide in the repository.	Advisor	**Planned**
M11	Final demo (CS 5002 expo)	Course	The finished product is shown to the instructor, advisor, and visitors.	Apr 16, 2027	Demo given on the expo day. Date to be fixed at M5 from the CS 5002 calendar.	Instructor and advisor	**Planned**

3. Table 1: Timeline (Assignment #6)

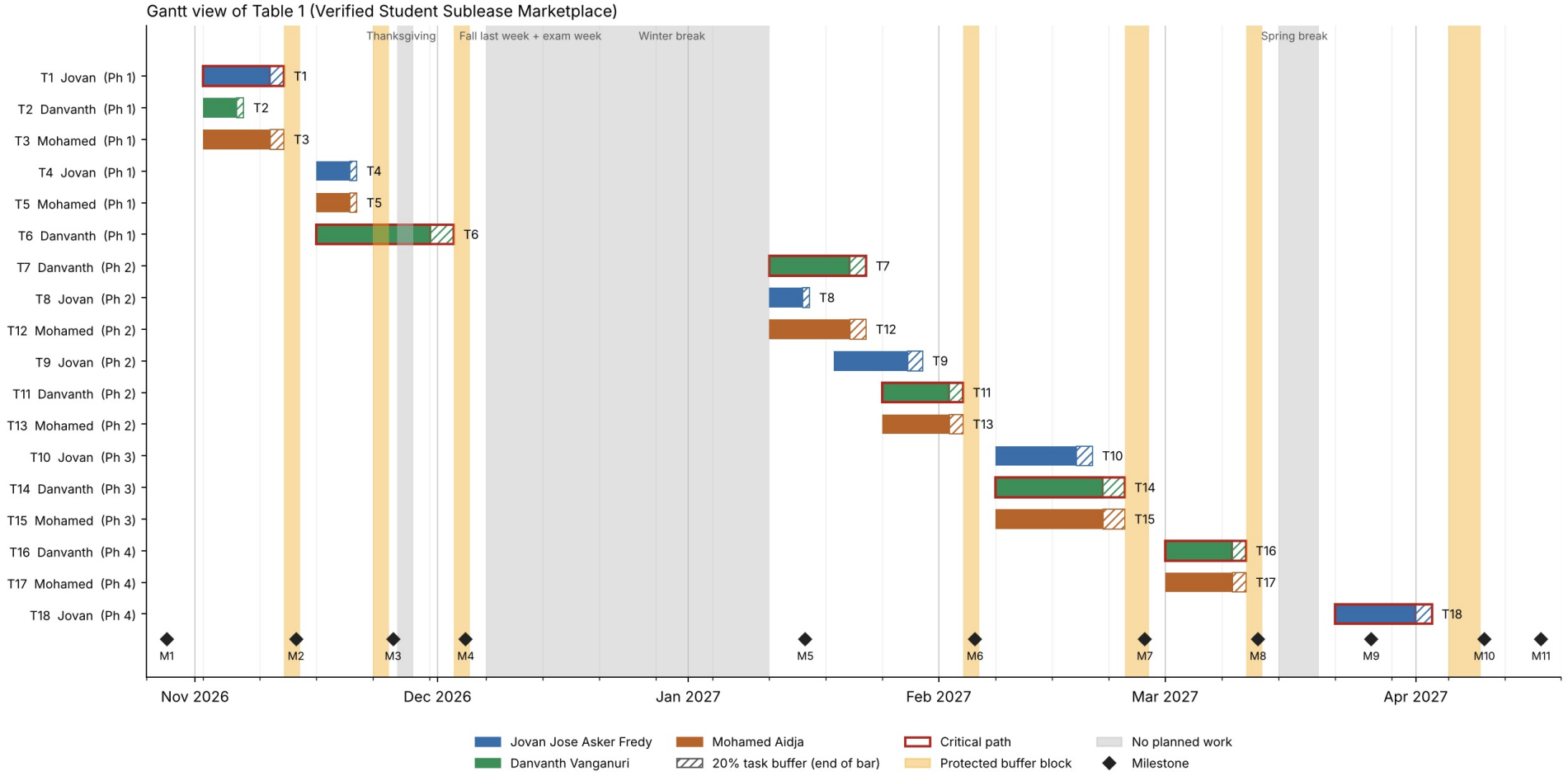
Rows are in date order. Gray rows are breaks with no work. Orange rows are protected buffer blocks. Yellow rows are milestones. **Hours** = expected (PERT) hours. **+20%** = task buffer already inside the start-to-finish dates. **Float** = spare work days before the next task or block needs it. **Red IDs (★)** are on the critical path: T1 → T6 → T7 → T11 → T14 → T16 → T18. They have zero float, so they only have their 20% buffer and the protected blocks. T3, T8, T13, T15, T17 also have zero float before their milestone block. **Risks** links each row to the Risk Register.

ID	Task statement / milestone	Owner	Start	Finish	Depends on	Hours	+20%	Float	Risks
M1	**Milestone: Assignment 8: Task list, milestones, timeline, effort matrix**	**Team**		**Oct 28, 2026**					
T1 ★	Develop the project skeleton, the CI pipeline, the G2 Configuration module, and the E3 database schema migrations.	Jovan	Nov 2, 2026	Nov 11, 2026	None	12.3	+2.5	0	RK4
T2	Research and obtain the E1 Email Delivery API account, and confirm its free-tier limit, webhook signing, and idempotency keys.	Danvanth	Nov 2, 2026	Nov 6, 2026	None	5.5	+1.1	3	RK1, RK5
T3	Design the C1 Web Client wireframes and the accessibility checklist for every user story screen.	Mohamed	Nov 2, 2026	Nov 11, 2026	None	10.3	+2.1	0	RK10
Buffer	Protected buffer block before M2. No new work. Fix, join parts, and test.	Team	Nov 12, 2026	Nov 13, 2026					RK3
M2	**Milestone: Development environment ready**	**Team**		**Nov 13, 2026**					
T4	Specify the C4 Compliance Screener phrase list v1, including the Ohio R.C. 4112 protected classes.	Jovan	Nov 16, 2026	Nov 20, 2026	None	8.3	+1.7	6	RK7
T5	Develop the G3 Logging and audit module (log, audit record, and the /healthz check).	Mohamed	Nov 16, 2026	Nov 20, 2026	T1	8.0	+1.6	6	-
T6 ★	Develop the G1 Authentication middleware and the signed session token.	Danvanth	Nov 16, 2026	Dec 2, 2026	T1	10.3	+2.1	0	RK3
Buffer	Protected buffer block before M3. No new work. Fix, join parts, and test.	Team	Nov 23, 2026	Nov 24, 2026					RK3
M3	**Milestone: Week 14 report (reproduces these tables)**	**Team**		**Nov 25, 2026**					
Break	*Thanksgiving: no planned work.*		*Nov 26, 2026*	*Nov 27, 2026*					*RK3*
Buffer	Protected buffer block before M4. No new work. Fix, join parts, and test.	Team	Dec 3, 2026	Dec 4, 2026					RK3
M4	**Milestone: Phase 1 done: sign-in base and screening rules**	**Team**		**Dec 4, 2026**					
Break	*Fall last week + exam week: no planned work.*		*Dec 7, 2026*	*Dec 18, 2026*					*RK3*
Break	*Winter break: no planned work.*		*Dec 19, 2026*	*Jan 10, 2027*					
T7 ★	Develop the C2 Identity Service for @mail.uc.edu login links, the 24-hour link expiry, and rate limits (I2, I10).	Danvanth	Jan 11, 2027	Jan 22, 2027	T2, T6	19.0	+3.8	0	RK1, RK2
T8	Develop the C4 Compliance Screener screen() function (I7) with unit tests for every phrase category.	Jovan	Jan 11, 2027	Jan 15, 2027	T1, T4	10.7	+2.1	0	RK7
T12	Develop the C1 Web Client screens for sign-in, create listing, and search (I1 to I4).	Mohamed	Jan 11, 2027	Jan 22, 2027	T3, T6	21.0	+4.2	8	RK8, RK10
M5	**Milestone: Spring re-kickoff and re-baseline**	**Team**		**Jan 15, 2027**					
Break	*MLK Day: no planned work.*		*Jan 18, 2027*	*Jan 18, 2027*					
T9	Develop the C3 Listing Service create, validate, hold, publish, and review endpoints (I3).	Jovan	Jan 19, 2027	Jan 29, 2027	T6, T8	20.7	+4.1	3	-
T11 ★	Develop the C8 Notification Service queue, the E1 send with retries, the status webhook, and the suppression list (I10 to I13).	Danvanth	Jan 25, 2027	Feb 3, 2027	T2, T7	18.3	+3.7	0	RK1, RK5
T13	Develop the C6 Messaging Service threads, messages, and alert limits (I5, I11).	Mohamed	Jan 25, 2027	Feb 3, 2027	T5, T6	16.7	+3.3	0	-
Buffer	Protected buffer block before M6. No new work. Fix, join parts, and test.	Team	Feb 4, 2027	Feb 5, 2027					RK3, RK8
M6	**Milestone: Walking skeleton**	**Team**		**Feb 5, 2027**					
T10	Develop the C3 outbox relay and the C5 Search & Match index and date-coverage search (I8, I4).	Jovan	Feb 8, 2027	Feb 19, 2027	T9	21.0	+4.2	2	RK4
T14 ★	Develop the C7 Approval Service proposals, decisions, unit registry seed, and daily reminder job (I6, I9, I12).	Danvanth	Feb 8, 2027	Feb 23, 2027	T9, T11	20.7	+4.1	0	RK6
T15	Develop the C1 Web Client screens for messages, approval requests, and the review queue (I5, I6).	Mohamed	Feb 8, 2027	Feb 23, 2027	T12, T13	19.0	+3.8	0	RK8
Buffer	Protected buffer block before M7. No new work. Fix, join parts, and test.	Team	Feb 24, 2027	Feb 26, 2027					RK3, RK8
M7	**Milestone: Core flows work end to end**	**Team**		**Feb 26, 2027**					
T16 ★	Test the timing budgets and error paths for AC-01.1, AC-02.1, and AC-02.2 with automated end-to-end tests.	Danvanth	Mar 1, 2027	Mar 10, 2027	T10, T11, T14	14.7	+2.9	0	RK3, RK4
T17	Validate US-04 accessibility with axe-core in CI and a manual NVDA and VoiceOver pass.	Mohamed	Mar 1, 2027	Mar 10, 2027	T12, T15	10.7	+2.1	0	RK10
Buffer	Protected buffer block before M8. No new work. Fix, join parts, and test.	Team	Mar 11, 2027	Mar 12, 2027					RK3
M8	**Milestone: Feature complete**	**Team**		**Mar 12, 2027**					
Break	*Spring break: no planned work.*		*Mar 15, 2027*	*Mar 19, 2027*					*RK9*

Break	*Spring break. No planned work.*		*Mar 15, 2027*	*Mar 19, 2027*					RK9
T18 ★	Validate the release candidate with a five-student user test, and document the results and the user guide.	Jovan	Mar 22, 2027	Apr 2, 2027	M8	14.3	+2.9	0	RK6, RK7
M9	**Milestone: User test sessions done**	**Team**		**Mar 26, 2027**					
Buffer	Protected buffer block before M10. No new work. Fix, join parts, and test.	Team	Apr 5, 2027	Apr 8, 2027					RK3
M10	**Milestone: Release candidate v1.0**	**Team**		**Apr 9, 2027**					
M11	**Milestone: Final demo (CS 5002 expo)**	**Team**		**Apr 16, 2027**					

Course milestones finished before this plan (M0a to M0e) are in Section 2.

Gantt view (supporting figure)



4. Table 2: Effort Matrix (Assignment #6)

Hours each member spends on each task. The owner's cell is shaded and marked ★. The owner's hours are the task's expected (PERT) hours. The other members' hours are code review, pairing, and helping join the parts. The owner always has the largest share.

ID	Module	Primary	Jovan Jose Asker Fredy	Danvanth Vanganuri	Mohamed Aidja	Row total	Owner share
T1	G2 Configuration; E3 Managed SQL DB	Jovan	★ **12.3**	1.5	1.5	**15.3**	80%
T2	E1 Email Delivery API (for C8)	Danvanth	0.5	★ **5.5**	0.5	**6.5**	85%
T3	C1 Web Client	Mohamed	1.5	1.5	★ **10.3**	**13.3**	77%
T4	C4 Compliance Screener	Jovan	★ **8.3**	1.0	1.0	**10.3**	81%
T5	G3 Logging and audit	Mohamed	1.0	1.0	★ **8.0**	**10.0**	80%
T6	G1 Authentication middleware	Danvanth	1.5	★ **10.3**	1.5	**13.3**	77%
T7	C2 Identity Service	Danvanth	1.5	★ **19.0**	2.0	**22.5**	84%
T8	C4 Compliance Screener	Jovan	★ **10.7**	1.0	0.5	**12.2**	88%
T9	C3 Listing Service	Jovan	★ **20.7**	1.5	2.0	**24.2**	86%
T10	C3 Listing Service; C5 Search & Match	Jovan	★ **21.0**	1.5	2.0	**24.5**	86%
T11	C8 Notification Service	Danvanth	1.5	★ **18.3**	1.0	**20.8**	88%
T12	C1 Web Client	Mohamed	2.0	2.0	★ **21.0**	**25.0**	84%
T13	C6 Messaging Service	Mohamed	1.5	1.5	★ **16.7**	**19.7**	85%
T14	C7 Approval Service	Danvanth	2.0	★ **20.7**	2.0	**24.7**	84%
T15	C1 Web Client	Mohamed	1.5	2.0	★ **19.0**	**22.5**	84%
T16	C3, C7, C8 (outbox and email path)	Danvanth	3.0	★ **14.7**	1.5	**19.2**	77%
T17	C1 Web Client	Mohamed	1.5	1.5	★ **10.7**	**13.7**	78%
T18	Whole system (C1 to C8)	Jovan	★ **14.3**	3.0	3.0	**20.3**	70%
Total			**106.3**	**107.5**	**104.2**	**318.0**	
Share			**33.4%**	**33.8%**	**32.8%**	**100%**	
	Of which: owned-task hours		87.3	88.5	85.7	261.5	
	Of which: Fall 2026 (Phase 1)		25.1	20.8	22.8	68.7	
	Of which: Spring 2027 (Phases 2 to 4)		81.2	86.7	81.4	249.3	

Balance check and rebalancing plan

Hours are even. The largest column (Danvanth Vanganuri, 107.5 h) is only 3.2% above the smallest (Mohamed Aidja, 104.2 h). Our limit is 10%.

Two things are not even, and here is what we will do:

- **Critical path load.** Danvanth Vanganuri owns 5 of the 7 critical-path tasks (T6, T7, T11, T14, T16). A slip there moves the whole plan. So Jovan Jose Asker Fredy is the backup for C8 and C7. He has 3 review hours on T16 and is free from Mar 1 to Mar 10 to take part of T16 if T14 is late.
- **Fall vs spring.** Fall work is small for everyone (Jovan 25.1 h, Danvanth 20.8 h, Mohamed 22.8 h). Most work is in spring. If a member goes on co-op in spring (RK2), we will move their spring tasks to the others at M5 and give that member review work and Phase 1 follow-ups.
- **Weekly check.** At each weekly meeting we log real hours next to this table. If any column gets more than 10% above the others, the member with the lowest total takes the next review or test task from them.

This table is the record the syllabus uses to adjust individual grades. We will update it with real hours in the Week 14 report and at M5.

5. Work Breakdown Structure (supporting artifact)

Level 1 is the phase. Level 2 is the task. Every task ID is the same as in the Task List, Table 1, and Table 2. Milestones close each phase.

WBS code	Phase / task	Module	Owner	Hours	Milestone
0	**Verified Student Sublease Marketplace**				
0.1	**Course deliverables (project management)**		**Team**		**M0a to M0e (done), M1, M3, M11**
1	**Phase 1: Foundation (Fall 2026, Nov 2 to Dec 4)**	**Shared setup, sign-in base, design inputs for spring.**		**54.7 h**	**M2, M4**
1.1	T1: Develop the project skeleton, the CI pipeline, the G2 Configuration module, and the E3 database schema migrations.	G2 Configuration; E3 Managed SQL DB	Jovan	12.3 h	M2
1.2	T2: Research and obtain the E1 Email Delivery API account, and confirm its free-tier limit, webhook signing, and idempotency keys.	E1 Email Delivery API (for C8)	Danvanth	5.5 h	M2
1.3	T3: Design the C1 Web Client wireframes and the accessibility checklist for every user story screen.	C1 Web Client	Mohamed	10.3 h	M2
1.4	T4: Specify the C4 Compliance Screener phrase list v1, including the Ohio R.C. 4112 protected classes.	C4 Compliance Screener	Jovan	8.3 h	M4
1.5	T5: Develop the G3 Logging and audit module (log, audit record, and the /healthz check).	G3 Logging and audit	Mohamed	8.0 h	M4
1.6	T6: Develop the G1 Authentication middleware and the signed session token.	G1 Authentication middleware	Danvanth	10.3 h	M4
2	**Phase 2: Core services (Spring 2027, Jan 11 to Feb 5)**	**Sign-in, listings, screening, email, first screens.**		**106.4 h**	**M5, M6**
2.1	T7: Develop the C2 Identity Service for @mail.uc.edu login links, the 24-hour link expiry, and rate limits (I2, I10).	C2 Identity Service	Danvanth	19.0 h	M6
2.2	T8: Develop the C4 Compliance Screener screen() function (I7) with unit tests for every phrase category.	C4 Compliance Screener	Jovan	10.7 h	M6
2.3	T9: Develop the C3 Listing Service create, validate, hold, publish, and review endpoints (I3).	C3 Listing Service	Jovan	20.7 h	M6
2.4	T11: Develop the C8 Notification Service queue, the E1 send with retries, the status webhook, and the suppression list (I10 to I13).	C8 Notification Service	Danvanth	18.3 h	M6
2.5	T12: Develop the C1 Web Client screens for sign-in, create listing, and search (I1 to I4).	C1 Web Client	Mohamed	21.0 h	M6
2.6	T13: Develop the C6 Messaging Service threads, messages, and alert limits (I5, I11).	C6 Messaging Service	Mohamed	16.7 h	M6
3	**Phase 3: Workflows (Spring 2027, Feb 8 to Feb 26)**	**Search, messaging, approval, and their screens.**		**60.7 h**	**M7**
3.1	T10: Develop the C3 outbox relay and the C5 Search & Match index and date-coverage search (I8, I4).	C3 Listing Service; C5 Search & Match	Jovan	21.0 h	M7
3.2	T14: Develop the C7 Approval Service proposals, decisions, unit registry seed, and daily reminder job (I6, I9, I12).	C7 Approval Service	Danvanth	20.7 h	M7
3.3	T15: Develop the C1 Web Client screens for messages, approval requests, and the review queue (I5, I6).	C1 Web Client	Mohamed	19.0 h	M7
4	**Phase 4: Test and release (Spring 2027, Mar 1 to Apr 16)**	**Timing tests, accessibility, user test, release.**		**39.7 h**	**M8, M9, M10**
4.1	T16: Test the timing budgets and error paths for AC-01.1, AC-02.1, and AC-02.2 with automated end-to-end tests.	C3, C7, C8 (outbox and email path)	Danvanth	14.7 h	M8
4.2	T17: Validate US-04 accessibility with axe-core in CI and a manual NVDA and VoiceOver pass.	C1 Web Client	Mohamed	10.7 h	M8
4.3	T18: Validate the release candidate with a five-student user test, and document the results and the user guide.	Whole system (C1 to C8)	Jovan	14.3 h	M10

6. Risk Register (supporting artifact)

Likelihood and impact are Low, Medium, or High. Strategy is avoid, reduce, transfer, or accept. **Where in Table 1** shows the task rows, buffer blocks, or milestones that carry the risk. High-impact schedule risks (RK1, RK2, RK4, RK8) are listed on those rows in Table 1.

ID	Category	Risk	L	I	Strategy	Mitigation	Owner	Trigger	Where in Table 1
RK1	Technical	UC Mail (Microsoft 365) puts our login emails in junk or blocks them, so students cannot sign in.	Medium	High	Reduce	T2 sends a real test email to a @mail.uc.edu inbox in week 1. We set SPF and DKIM on the sending domain. We keep a second provider as a backup.	Danvanth	A test email lands in junk or bounces.	T2, T7, T11; checked at M2
RK2	Team	A member goes on co-op in Spring 2027 and has less time or works off campus.	Medium	High	Avoid	Each member confirms spring status by M5 (Jan 15). If someone is on co-op, we move their spring tasks to the others and they take review work. All work is in the repository, so it can be done remotely.	Jovan	Any member accepts a Spring 2027 co-op.	M5 re-baseline; T7
RK3	Team	Exams, Thanksgiving, and other courses slow us down.	High	Medium	Reduce	No tasks during fall exam week or winter break. Every task has 20% buffer. Each major milestone has a protected buffer block.	Mohamed	A task is more than 25% behind at the weekly check.	Blocked rows and buffer rows in Table 1; T6, T16
RK4	Technical	The free hosting tier is missing the btree_gist extension, or the Render instance sleeps and the outbox relay stops. Search and email then miss AC-01.1.	Medium	High	Reduce	T1 tests both before M2. If the instance sleeps, we pay for the always-on tier (inside the about $7 per month budget, NFR-07).	Jovan	Extension not available, or a gap over 60 s between relay polls in the logs.	T1, T10, T16; checked at M2
RK5	External	The E1 email provider changes its free limit or price.	Low	Medium	Transfer	The provider is responsible for delivery under its terms. C8 keeps the provider behind one interface (I13), so we can switch providers in about a day.	Danvanth	Monthly use goes over 70% of the free limit, or the terms change.	T2, T11
RK6	External	We cannot find a real property manager to test US-03 (approval).	Medium	Medium	Accept	We seed test property-manager accounts that the team controls (D-09, GQ-04). We ask the advisor for a real contact before M7.	Danvanth	No property-manager contact by Feb 26, 2027.	T14, T18
RK7	Technical	The Fair Housing phrase list misses a phrase or flags too many fair listings.	Medium	Medium	Reduce	Advisor reviews phrase list v1 at M4. Flagged listings go to a person for review, never straight out. The user test checks false flags.	Jovan	More than 10% false flags in the user test, or a missed phrase is found.	T4, T8, T18; checked at M4
RK8	Technical	The C1 screens and the backend APIs stop matching, because screens are built against mock data first.	Medium	High	Reduce	Shared zod schemas are the contract. Contract tests run in CI. The protected blocks before M6 and M7 are used for joining the parts.	Mohamed	A contract test fails in CI.	T12, T15; buffer before M6 and M7
RK9	External	The CS 5002 calendar and spring break dates are not published yet, so spring dates may move.	High	Low	Accept	We re-baseline Table 1 at M5 when the calendar is posted.	Jovan	The CS 5002 calendar is posted.	M5, M11
RK10	Technical	Accessibility problems (US-04) are found late.	Medium	Medium	Reduce	The accessibility checklist is part of T3. axe-core runs in CI from T12, not only in T17.	Mohamed	axe-core reports any violation on main.	T3, T12, T17

10 risks. Categories used: External, Team, Technical. Strategies used: Accept, Avoid, Reduce, Transfer.

7. Consistency check

The PDF, Tasklist.md, and the CSV files are all made from one file (Planning/plan.py). So every task has the same ID, name, and owner everywhere. The check below is run by the build script.

ID	Owner	Task List	Tasklist.md	WBS	Table 1	Table 2	Owner has largest share
T1	Jovan	✓	✓	✓	✓	✓	✓
T2	Danvanth	✓	✓	✓	✓	✓	✓
T3	Mohamed	✓	✓	✓	✓	✓	✓
T4	Jovan	✓	✓	✓	✓	✓	✓
T5	Mohamed	✓	✓	✓	✓	✓	✓
T6	Danvanth	✓	✓	✓	✓	✓	✓
T7	Danvanth	✓	✓	✓	✓	✓	✓
T8	Jovan	✓	✓	✓	✓	✓	✓
T9	Jovan	✓	✓	✓	✓	✓	✓
T10	Jovan	✓	✓	✓	✓	✓	✓
T11	Danvanth	✓	✓	✓	✓	✓	✓

T12	Mohamed	✓	✓	✓	✓	✓	✓
T13	Mohamed	✓	✓	✓	✓	✓	✓
T14	Danvanth	✓	✓	✓	✓	✓	✓
T15	Mohamed	✓	✓	✓	✓	✓	✓
T16	Danvanth	✓	✓	✓	✓	✓	✓
T17	Mohamed	✓	✓	✓	✓	✓	✓
T18	Jovan	✓	✓	✓	✓	✓	✓